



Post-traumatic Stress Disorder

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Introduction

In the Diagnostic and Statistical Manual of Mental Disorders 5th edition (DSM-5), post-traumatic stress disorder (PTSD) is classified with other trauma- and stressor-related disorders that are diagnosed as occurring due to exposure to a traumatic or stressful event.^[1] Other conditions in this classification include reactive attachment disorder, acute stress disorder and adjustment disorders.

PTSD is characterized by significant distress or impairment in functioning of at least 1 month's duration in response to a traumatic event described as exposure to actual or threatened death, serious injury or sexual violence, learning that this (event) occurred to a close relative or close friend, or experiencing repeated or extreme exposure to aversive details of the event.^[1]

Symptoms have been divided into 4 main categories and are included in [Table 1](#). To make the diagnosis, the individual must present with at least 1 intrusive re-experiencing symptom where the individual relives the traumatic event in intrusive thoughts and recollections, 1 or more symptoms of avoidance, at least 2 negative cognition and mood symptoms, and at least 2 arousal and reactivity symptoms. Finally, to make the diagnosis, the individual exposed to the traumatic event must experience significant impairment in social and/or occupational functioning as a result of their symptoms for greater than 1 month. Children 6 years of age or younger with the disorder are now diagnosed using a separate set of criteria.^[1] This chapter focuses on PTSD in the adult patient population.

Goals of Therapy

- Eliminate or decrease frequency and severity of PTSD symptoms
- Eliminate or decrease PTSD-based disability
- Prevent relapse

- Treat comorbid conditions

Investigations

The assessment for PTSD requires a detailed exploration of the nature and circumstances of the traumatic event(s) and the subsequent effect on the patient. It is important to be sensitive to and patient with the person's potential inability to recall these details without becoming overwhelmed. It may take a number of sessions to allow for a complete understanding of the circumstances of the traumatic experience(s) and to establish a strong therapeutic alliance in order to enable the patient to willingly/comfortably undergo therapy.

- Thorough history with attention to:
 - nature of symptoms and onset
 - nature and extent of disability
 - presence of comorbid medical or psychiatric conditions (see [Initial Management](#))
- Criteria to assist in obtaining an accurate diagnosis (see [Table 1](#)). In addition to DSM-5 diagnostic criteria, the following screening tools are available:
 - [Primary Care PTSD Screen for DSM-5 or PC-PTSD-5](#)
 - a quick 5-item screening tool designed for use in primary care
 - [PTSD Checklist for DSM-5 or PCL-5](#)
 - a self-administered tool to assist in monitoring symptoms during treatment; also useful for screening and provisional diagnosis
 - [Clinician-Administered PTSD Scale for DSM-5 or CAPS-5](#)
 - considered the gold standard for diagnosis; can take 20–40 minutes to complete
- Physical examination to exclude endocrine or cardiac disorders and to look for signs of substance misuse. Beyond ruling out the possibility that symptoms are due to a medical condition, it is important to keep in mind the increased rate of medical illness in those with psychiatric disorders.^[2]
 - [NIDA Quick Screen](#) allows primary care practitioners to screen for drug and alcohol use
- Laboratory tests:
 - CBC, electrolytes, glucose, liver function tests including GGT (to screen for alcohol use), thyroid function tests, renal function tests, serum ferritin, vitamin B₁₂, ECG

Table 1: Diagnosis Criteria for PTSD^[7]

Symptoms	Examples
Intrusion (repeated intrusion symptoms after the traumatic event)	Intrusive memories, distressing dreams, dissociative symptoms such as flashbacks, intense distress cued by aspects of the trauma
Avoidance	Attempts to avoid memories, thoughts, feelings or external reminders associated with the triggering event
Negative cognition and mood	A broad category that may include a distorted sense of blame of self or others, inability to recall aspects of the traumatic event, loss of interest in activities
Arousal and reactivity	Hypervigilance, sleep disturbances, self-destructive behaviour, irritability, verbal or physical aggression, recklessness

Therapeutic Choices

An algorithm of the management of PTSD is provided in [Figure 1](#).

Initial Management

Most people who experience trauma do not go on to develop either acute stress disorder or PTSD. In the first 4 weeks following a traumatic event, psychotherapy or pharmacotherapy, including benzodiazepines, is usually not required. Psychological debriefing, a short intervention delivered in group settings during the first 2 weeks following a traumatic event, may impede natural recovery and is not usually recommended.^[3] During this initial phase, the goal is to help patients to understand that acute responses to trauma are common and often short term. Allow them to use their natural resilience and usual emotional supports to recover.

There is little evidence to show that medication taken immediately after a traumatic event can prevent the onset of PTSD.^[4] For those individuals who are overwhelmed and have impaired functioning, psychotherapy of a supportive nature or medication may need to be instituted within the first 4 weeks following the traumatic event.

Treatment of PTSD may vary depending on the severity and chronicity of trauma symptoms and the presence of comorbid disorders. Most often a phase-oriented approach to treatment is utilized, which

includes behavioural stabilization, psychoeducation, anxiety management, trauma-focused psychotherapy, relapse prevention and aftercare.^[5] Treat comorbid mood disorders as the primary condition if they are severe enough to interfere with PTSD treatment (see [Depression](#)). It is important to manage comorbid chronic pain and sleep disturbances in patients with PTSD (see [Neuropathic Pain](#) and [Insomnia](#)). Patients with PTSD often attempt to self-medicate and can have substance use disorders and/or other behavioural addictions (see [Alcohol-Related Disorders](#), [Opioid-Related Disorders](#) and [Tobacco Use Disorder: Smoking Cessation](#)); referral to a specialized program may be necessary.

Trauma-focused psychotherapy is recommended as first-line treatment before initiating pharmacotherapy.^{[6][7][8][9][10][11]} Although trauma-focused psychotherapy has stronger evidence,^[12] pharmacotherapy is also recommended as first-line treatment when trauma-focused psychotherapy is not available or not preferred, or if stabilization is required;^{[7][8][9][13]} pharmacotherapy may also be more effective than psychotherapy in military or combat-related PTSD.^[14] PTSD guidelines were unable to make a clear recommendation for combining psychotherapy and pharmacotherapy, but a study published after the guidelines showed no significant difference between either approach alone or the combination.^[15] Patients with PTSD should be offered a choice that best fits with their circumstances, available resources and preferences, as improved adherence and outcomes have been observed when patients receive their preferred treatment.^[16]

Nonpharmacologic Choices

Several psychological approaches have been investigated in the treatment of PTSD and acute stress disorder. The primary goal of **psychotherapy** is to reduce symptom severity and improve quality of life and functioning in social and occupational areas. **Cognitive behavioural therapy** (CBT) is effective for reducing symptoms of PTSD.^[3] Evidence shows that therapies focused on the trauma, such as **prolonged exposure** (PE), **cognitive processing therapy** (CPT),^{[8][9][17]} and **eye movement desensitization and reprocessing therapy** (EMDR) are more effective than more general approaches (see [Table 2](#)).^{[6][10]} Supportive psychotherapy that does not deal with the trauma directly is no more effective than being on a waiting list.^[18] There is also emerging evidence on the benefits of using adaptive disclosure intervention to address moral injury, especially in the context of military-related PTSD.^{[19][20]} Moral injury describes the distress resulting from an event that violates deeply held moral beliefs. While diagnostically separate from PTSD, moral injury is often concurrent with PTSD and may exacerbate symptoms.

Regardless of the psychotherapeutic treatment modality, prior behavioural stabilization is critical, as some comorbid conditions (e.g., suicidality, self injurious behaviour or psychosis) may preclude the introduction of treatment. Initiating trauma-focused psychotherapy prior to stabilization may exacerbate pre-existing comorbid symptoms of depression and substance abuse.

In patients who do not require stabilization and are unable or delayed in accessing in-person or virtual psychotherapy, internet-based CBT can be considered.^[21]

Lifestyle modification is an emerging nonpharmacologic avenue. Exercise augmentation to usual care has been shown to reduce symptoms of PTSD and comorbid depression.^{[22][23]}

Table 2: Psychotherapy Options for PTSD^[24]

Psychotherapy	Description
Cognitive Behavioural Therapy (CBT)	The patient is guided to modify beliefs and behaviours with the aim of influencing disturbed emotions.
Prolonged Exposure (PE)	The patient confronts the trauma imaginally or in vivo during planned treatment sessions
Cognitive Processing Therapy (CPT)	The patient focuses on the appraisal of the traumatic event and the emotions and beliefs resulting from the event
Eye Movement Desensitization and Reprocessing Therapy (EMDR)	The patient is instructed to imagine the traumatic memories and associated negative cognitions (e.g., guilt, shame) while visually focusing on the movement of the clinician's finger ^[25]

Pharmacologic Choices

The pharmacologic management of PTSD depends on the nature, severity, and frequency of the trauma and requires a multimodal treatment program. Short-term treatment with medication may benefit patients who do not have access to trauma-focused therapy.

SSRIs and SNRIs have been shown to reduce the severity of all the symptom clusters of PTSD. **Fluoxetine**, **paroxetine**,^[26] **sertraline**^[27] and **venlafaxine**^[28] are first-line options. Other antidepressants with evidence of efficacy that are considered second-line choices include **fluvoxamine**,^[29] **moclobemide**^[30] and **phenelzine**.^{[31][32]} **Mirtazapine** is also considered a second-line option,^{[33][34]} particularly for comorbid depression, insomnia and loss of appetite/weight. However, it may not be effective enough as monotherapy^[35] and may be more useful as adjunctive therapy. A meta-analysis of relapse prevention trials in patients with remitted anxiety disorders, including PTSD, treated with antidepressants found a clear benefit of continuing treatment for up to 1 year for both relapse rate and time to relapse.^[36]

Second-generation **antipsychotics** such as **aripiprazole**, **quetiapine** and **olanzapine** may be effective in augmenting the effects of antidepressants in some patients who do not respond to first-line PTSD treatment and may benefit those patients with prominent hyperarousal or re-experiencing symptoms.^{[11][37]} ^[38] Monotherapy with antipsychotics is not routinely recommended.^[10] Patients on antipsychotics should be monitored regularly for metabolic adverse effects such as weight gain, glucose abnormalities and dyslipidemias; for specific recommendations, see [Schizophrenia and Related Psychotic Disorders](#), [Table 5](#).

Monotherapy with **benzodiazepines** is not recommended due to their lack of efficacy and the high rates of substance use disorders in patients with PTSD.^{[10][39]} Adjunctive benzodiazepines are also generally not recommended, although they may be used in patients receiving evidence-based treatment for PTSD, for short-term relief of acute anxiety and/or insomnia.^[9] Benzodiazepines are associated with rebound insomnia upon discontinuation, especially after long-term use.^[40] Avoid using benzodiazepines in patients with PTSD who have comorbid substance use disorders.

Sleep disturbances are common in patients with PTSD. Although evidence from clinical trials is limited in this patient population, several PTSD practice guidelines support the use of **trazodone**^[41] in the management of insomnia that has not responded to nonpharmacologic therapy (e.g., sleep hygiene, CBT-insomnia) or treatment for comorbid conditions. **Prazosin** is an alpha₁-adrenergic antagonist that reduces sympathetic outflow in the brain. It is often used in clinical practice for the treatment of PTSD-associated nightmares, despite mixed evidence from small clinical trials.^{[42][43][44]} A more robust meta-analysis revealed that prazosin was helpful for nightmares, reduced PTSD-related hyperarousal, and improved total sleep time and quality.^[45] Prazosin may be more effective in patients with adrenergic activation or unstable PTSD (e.g., hypertension, anxiety, agitation, self-medicating with alcohol); further studies to identify a subgroup of PTSD patients responsive to prazosin are required.

See [Table 3](#) for more information on medications used in the management of PTSD.

Complementary Therapy Choices

In more recent years, **cannabis** for medical purposes has been proposed for the treatment of PTSD. Despite its use, there is little research on the efficacy for treatment of mental health conditions. Cannabis may increase risk of comorbidities, particularly psychosis and suicidal ideation.^[46] Due to limited evidence for the treatment of PTSD with plant-based cannabis or cannabis derivatives, and known risks of cannabis, it is not recommended for the treatment of PTSD.^{[10][47]}

Choices during Pregnancy and Breastfeeding

PTSD and Pregnancy

Patients who have encountered trauma during their lives may experience a worsening or relapse of PTSD symptoms during pregnancy.

Many people with PTSD do not bring their distress to the attention of their caregivers. If possible, screen patients for the presence of PTSD or anxiety symptoms prior to conception. Screening can be repeated during the pregnancy and particularly postpartum. It is imperative to screen for the presence of mood symptoms and suicidality in a patient who is suffering from marked psychological distress related to pregnancy and breastfeeding.

It may be necessary to refer patients experiencing severe PTSD symptoms during pregnancy or postpartum to a psychiatrist. Mental health programs for patients during pregnancy are usually available in major centres and these programs customarily respond quickly to consultation requests.

Treatment for PTSD or other psychiatric disorders that are producing significant distress or interfering with functioning can be offered to patients before conception.

Management during Pregnancy

Evidence-based psychotherapies such as CBT can be administered without restriction throughout pregnancy. There is good evidence to show that psychological treatments can have beneficial effects for more than half of the patients who remain with a treatment program. Therapies based in meditative or relaxation techniques may be more acceptable to the patient than pharmacologic approaches.^[48]

If PTSD symptoms are severe and producing significant impairment, medications can be appropriate and effective.^[49] **SSRIs** are the mainstay of pharmacologic therapy for PTSD during pregnancy. The use of **SSRIs** may be warranted in patients with severe symptoms that could affect fetal or maternal safety or health. In general, use the lowest effective dose for the shortest time necessary. General principles for management of depression during pregnancy are applicable to the management of PTSD. The Canadian Network for Mood and Anxiety Treatments (CANMAT) recommends **citalopram**, **escitalopram** and **sertraline** as first-line options in pregnancy.^[50]

There have been reports of a slightly higher (but still low) risk of congenital abnormalities involving the heart or cleft lip/palate, particularly with the use of paroxetine.^[51] Use of these drugs in the third trimester may be associated with neonatal withdrawal symptoms such as tremors, increased muscle tone, feeding or digestive problems, or respiratory distress. Whether benzodiazepines confer an increased risk of congenital malformations is controversial.^{[52][53][54]}

See also [Depression](#) and [Schizophrenia and Related Psychotic Disorders](#) for information on the use of antidepressants and antipsychotics during pregnancy and breastfeeding.

PTSD and Breastfeeding

In the postpartum period, severe distress can impede the patient's sleep and erode confidence in caring for the child.

As in pregnancy, nonpharmacologic options should be used whenever possible in the postpartum period, particularly in breastfeeding patients. If drug therapy is necessary, consider **sertraline**, **escitalopram** or **citalopram**, since they appear in low concentrations in breast milk.^{[50][55]}

A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#), and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

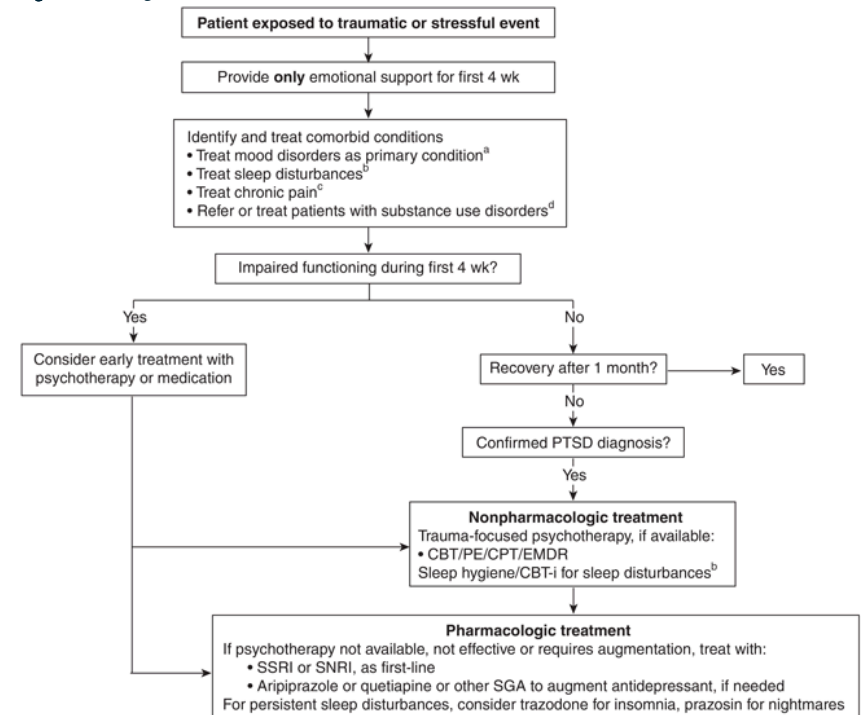
Therapeutic Tips

- Trauma-focused psychotherapy is recommended as first-line treatment for PTSD whenever possible.
- Pharmacotherapy is also recommended as first-line, especially in the presence of comorbidity.

- When selecting treatment, always consider patient preferences and expectations, as this can directly impact outcomes.^{[16][56]}
- Assess a drug's effectiveness after a trial of an adequate dosage taken for an adequate length of time (4–6 wk). The [PTSD Checklist for DSM-5 or PCL-5](#) and the [Clinician-Administered PTSD Scale for DSM-5 or CAPS-5](#) may be useful monitoring tools for the ongoing management of PTSD.
- If an antidepressant is not effective, switch to a second agent from the same or a different drug class.
- If a second antidepressant is not effective, switch to an agent from a different drug class or use augmentation strategies.
- Families of persons with PTSD can be significantly affected; refer partners, children, or other family members to counsellors when indicated.

Algorithms

Figure 1: Management of Post-traumatic Stress Disorder



[a] See [Depression](#).
[b] See [Insomnia](#).
[c] See [Neuropathic Pain](#).
[d] See [Alcohol-Related Disorders](#), [Opioid-Related Disorders](#) and [Tobacco Use Disorder: Smoking Cessation](#).

Abbreviations: CBT = cognitive behavioural therapy; CBT-i = cognitive behavioural therapy for insomnia; CPT = cognitive processing therapy; EMDR = eye movement and desensitization reprocessing; PE = prolonged exposure; PTSD = post-traumatic stress disorder; SGA = second-generation antipsychotic; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor

Drug Table

Table 3: Drugs Used in the Management of Post-traumatic Stress Disorder

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Alpha ₁ -adrenergic Antagonists				
prazosin generics	Initial: 1 mg QHS PO Increase by 1–2 mg QHS PO until effective dose reached (range 3–15 mg QHS PO)	Orthostatic hypotension (particularly following first dose), dizziness.	Increased risk of orthostatic hypotension with other antihypertensives or PDE5 inhibitors (sildenafil, tadalafil, vardenafil).	Used to improve sleep and/or reduce nightmares in patients with PTSD . Current or past use of alpha ₁ -adrenergic antagonists increases the risk of intraoperative floppy iris syndrome during cataract surgery; discontinuing the drug preoperatively does not eliminate the risk. Inform the ophthalmologist if prazosin has been used at any

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
Drug Class: Antipsychotics, second-generation				
aripiprazole Abilify , generics	Initial: 2–5 mg daily PO Titrate gradually to effect, in maximum increments of 5 mg/day every 7 days Maximum: 15 mg daily PO	EPS (akathisia, parkinsonism), dizziness, orthostatic hypotension, headache, GI complaints, nasopharyngitis, tremor, sedation, insomnia.	Strong inducers of CYP2D6 or CYP3A4 (e.g., carbamazepine, phenytoin, rifampin) can decrease aripiprazole levels substantially. Strong inhibitors of CYP2D6 or CYP3A4 (e.g., ketoconazole, quinidine, fluoxetine, paroxetine) can increase aripiprazole levels substantially.	time prior to cataract surgery. Used as augmentation therapy with first-line agents in PTSD. Advise patients about antipsychotic-associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Associated with a lower likelihood of weight gain and anticholinergic effects than other SGAs . Dose in the morning to reduce impact of activating adverse effects; activation usually occurs with higher doses and may be alleviated by lowering the dose.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
olanzapine Zyprexa , Zyprexa Zydis, Olanzapine, Olanzapine ODT, other generics < \$25	Initial: 2.5 mg daily PO Titrate gradually to desired effect, usually 2.5–5 mg daily PO. May need to increase to a maximum of 10 mg/day PO	Weight gain, dizziness, sedation, anticholinergic effects, hepatic aminotransferase elevation, orthostatic hypotension, increased risk of diabetes and dyslipidemia, extrapyramidal effects (especially akathisia). Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) is a rare but very serious adverse effect. See Drug- Induced Skin Reactions for more information regarding DRESS.	Sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP1A2 or CYP2D6 such as diltiazem, fluvoxamine or paroxetine may increase olanzapine levels; inducers of CYP1A2 or CYP3A4 such as barbiturates, carbamazepine, phenytoin or rifampin may decrease olanzapine levels. Cigarette smoking induces CYP1A2 and may necessitate a higher dose of olanzapine. Conversely, a reduction of the dose may be required if the patient quits smoking.	Used as augmentation therapy with first- line agents in PTSD . Advise patients about antipsychotic- associated body temperature dysregulation and prevention of heat stroke, e.g., hydration, sun protection. Associated with a higher likelihood of sedation, weight gain and anticholinergic effects than other SGAs .
quetiapine extended- release Seroquel XR, generics \$25–50	Initial: 50 mg daily PO for 2 days, then 150 mg daily; if necessary, can increase to 300 mg on or after day 4 Usual: 150– 300 mg/day	Sedation, dizziness, weight gain, orthostatic hypotension, hepatic transaminase elevation, headache, anticholinergic effects, increased	Additive sedation with CNS depressants; may potentiate antihypertensive drug effects; inhibitors of CYP3A4 such as clarithromycin, erythromycin,	Used as augmentation therapy with first- line agents in PTSD. Associated with a higher likelihood of sedation than other SGAs .

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
		risk of diabetes and dyslipidemia, possible increased risk of cataracts; may reduce thyroid hormone levels.	grapefruit juice, ketoconazole or prednisone may increase quetiapine levels; inducers of CYP3A4 such as carbamazepine, phenytoin or rifampin may reduce quetiapine levels.	
Drug Class: Dual Action Antidepressants				
mirtazapine  Remeron, Remeron RD, Mirtazapine, other generics < \$25	Initial: 15–30 mg QHS PO Usual: 30– 45 mg/day	Somnolence, increased appetite/weight gain, dizziness.	Do not use with MAOIs ; additive sedation with other CNS depressants such as alcohol, benzodiazepines; substrate of CYP1A2 , 2D6 and 3A4—caution with inhibitors or inducers of these isoenzymes.	May be most beneficial in patients with comorbid insomnia, depression, low appetite and/or weight loss. Most sedating at lower doses. Initiate at lower doses (15 mg/day) if targeting insomnia; start at higher doses (30 mg/day) if targeting depression.
Drug Class: Irreversible Monoamine Oxidase Inhibitors				
phenelzine Nardil \$50–75	Initial: 15– 30 mg/day PO Usual: 45– 90 mg/day PO	Insomnia, dizziness, orthostatic hypotension, edema, sexual dysfunction.	Concurrent use with sympathomimetic agents, tyramine or levodopa may result in hypertensive crisis; do not use	Stringent dietary restrictions of tyramine- containing foods is necessary.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
			with serotonergic drugs such as SSRIs , SNRIs , TCAs , meperidine, tryptophan (increased risk of serotonin syndrome).	

Drug Class: Reversible Monoamine Oxidase-A Inhibitors

moclobemide Manerix, generics \$25–50	Initial: 200– 300 mg/day PO Usual: 450– 600 mg/day PO	Nausea, insomnia.	Concurrent use with sympathomimetic agents, tyramine or levodopa may result in hypertensive crisis; do not use with serotonergic drugs such as SSRIs , SNRIs , TCAs , meperidine, tryptophan (increased risk of serotonin syndrome).	Dietary restrictions are not required at usual doses (<600 mg/day)
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Drug Class: Selective Serotonin Reuptake Inhibitors (SSRIs)

fluvoxamine Luvox, generics \$25–50	Initial: 50– 100 mg/day PO Usual: 100– 300 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with	May take 2–3 months to achieve maximum effect. Discontinue gradually.
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		Fluvoxamine: somnolence and nausea are common.	other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Fluvoxamine: potent inhibitor of CYP1A2 .	
fluoxetine Prozac, Odan- Fluoxetine, other generics < \$25	Initial: 10– 20 mg/day PO Usual: 20– 40 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Fluoxetine: anxiety upon initiation is common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Fluoxetine: potent inhibitor of CYP2D6 .	May take 2–3 months to achieve maximum effect. Longer half-life (4–6 days) allows for a more rapid discontinuation than other SSRIs .

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
paroxetine immediate-release Paxil , Paroxetine, other generics < \$25	Initial: 10–20 mg/day PO Usual: 20–40 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia. Paroxetine: anticholinergic effects (dry mouth, constipation), somnolence are common.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Paroxetine: potent CYP2D6 inhibitor.	May take 2–3 months to achieve maximum effect. Discontinue gradually.
paroxetine controlled-release Paxil CR \$50–75	Initial: 12.5–25 mg/day PO Usual: 25–37.5 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction,	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if	May take 2–3 months to achieve maximum effect. Discontinue gradually.

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
		weight gain, SIADH with hyponatremia. Paroxetine: anticholinergic effects (dry mouth, constipation), somnolence are common.	combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Paroxetine: potent CYP2D6 inhibitor.	

Drug/Cost ^[a]	Dosage	Adverse Effects	Drug Interactions	Comments
sertraline Zoloft , generics < \$25	Initial: 25–50 mg/day PO Usual: 50–200 mg/day PO	Upon initiation: GI upset, anorexia, dry mouth, diaphoresis, headache, dizziness, insomnia, somnolence, anxiety, agitation, tremor. Usually resolve after 2 wk of therapy. Others: sexual dysfunction, weight gain, SIADH with hyponatremia.	SSRIs are all substrates of various CYP enzymes and subject to drug interactions. Concurrent use with MAOIs , linezolid and methylene blue is contraindicated due to increased risk of serotonin syndrome. Use caution if combining with other serotonergic agents. Increased risk of GI bleeding with NSAIDs , antiplatelet agents. Avoid concurrent use with drugs associated with prolonged QTc interval/torsades de pointes. Sertraline: weak CYP2D6 inhibitor.	May take 2–3 months to achieve maximum effect. Discontinue gradually.

Drug Class: Serotonin Modulators

trazodone generics < \$25	Initial: 25–50 mg QHS PO May increase gradually up to 100 mg QHS PO as needed/tolerated	Drowsiness, orthostatic hypotension, nausea, vomiting, headache, dry mouth, priapism (rare).	Toxicity may be increased by inhibitors of CYP3A4 such as clarithromycin, erythromycin, grapefruit juice, ketoconazole. Effectiveness may be decreased by inducers of CYP3A4	For management of insomnia.
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			such as carbamazepine, phenytoin, rifampin.	
Drug Class: Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs)				
desvenlafaxine  Pristiq, generics \$50–75	Initial/usual: 50 mg daily PO Maximum: 100 mg daily PO	Nausea, sleep disturbance, drowsiness, nervousness, dizziness, dry mouth.	Do not use with MAOIs . Substrate of CYP3A4 ; caution with inducers or inhibitors of this isoenzyme.	May take 2–3 months for maximum effect. Discontinue gradually.
venlafaxine  extended-release Effexor XR, generics < \$25	Initial: 37.5–75 mg/day PO Usual: 75–225 mg/day PO	Nausea, insomnia, dizziness, asthenia.	Do not use with MAOIs . Substrate of CYP2D6 and CYP3A4 ; caution with inducers or inhibitors of these isoenzymes.	May take 2–3 months for maximum effect. Discontinue gradually.

^[a] Cost of 30-day supply of usual mean dosage; includes drug cost only.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).
Abbreviations: CNS = central nervous system; CYP = cytochrome P450; EPS = extrapyramidal symptoms; GI = gastrointestinal; MAOI = monoamine oxidase inhibitor; NSAID = nonsteroidal anti-inflammatory drug; PDE5 = phosphodiesterase type 5; PTSD = post-traumatic stress disorder; SIADH = syndrome of inappropriate antidiuretic hormone; SGA = second-generation antipsychotic; SNRI = serotonin-norepinephrine reuptake inhibitor; SSRI = selective serotonin reuptake inhibitor; TCA = tricyclic antidepressant

Suggested Readings

[Katzman MA, Bleau P, Blier P et al. Canadian clinical practice guidelines for the management of anxiety, posttraumatic stress and obsessive-compulsive disorders. *BMC Psychiatry* 2014;14\(Suppl 1\):S1.](#)

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